

CSE370: Database Systems

Lecture 7: Relational Algebra



Inspiring Excellence

BACK TO SQL

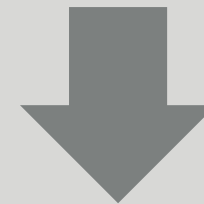
How does a computer understand SQL queries? 🤔

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

-- QUERY

SELECT * FROM STUDENT
WHERE Dept='CSE' AND Attn_Rate > 0.7;

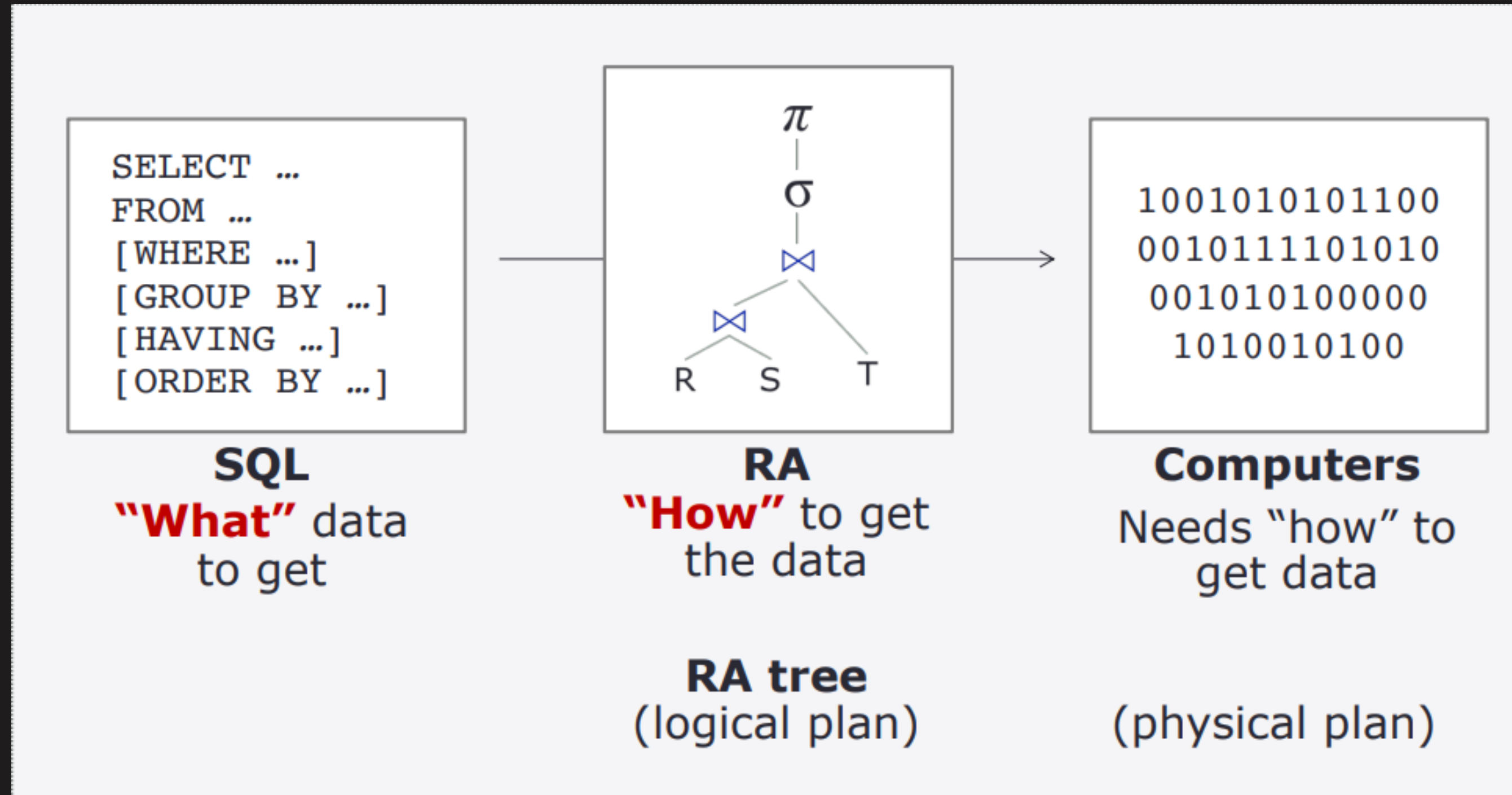


SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0

- When a RDBMS processes a query, it translates it into a **RA tree**
- After some optimization, the **RA tree** is converted into instructions

DBMS AND QUERY PROCESSORS

RDBMS



RELATIONAL ALGEBRA (RA)



Overview

- The basic set of operations for the relational model
- Enable a user to specify basic retrieval requests (or queries)
- The result of an algebra operation is a *new relation*

Unary Relational Operations

Take **one** relation, return a new relation

- ✿ SELECT (symbol: σ (sigma))
- ✿ PROJECT (symbol: π (pi))
- ✿ RENAME (symbol: ρ (rho))

Additional Expressiveness

Assignment (symbol: $\leftarrow$)

Aggregate Functions (symbol: G)

Binary Relational Operations

Take **two** relation, return a new relation

- ✿ Union ($\cup$)
- ✿ Intersection ($\cap$)
- ✿ Difference ($-$)
- ✿ Cartesian Product ($\times$)
- ✿ Join ($\bowtie$ (theta))

SELECTION (σ)

$\sigma_c(R)$ - where C is a set of conditions that involve attribute(s) of R

- ✱ Unary operation – take one operand
- ✱ Return all tuples that satisfy a condition (filter tuples)
- ✱ Conditions can be =, <, ≤, >, ≥, ≠ and combined with AND, OR, NOT

Find students from CSE dept with attn_rate above 0.7

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

-- EQUIVALENT QUERY
SELECT * FROM STUDENT
WHERE Dept='CSE' AND Attn_Rate > 0.7;

$\sigma_{\text{Dept}='CSE' \text{ AND Attn_rate} > 0.7}$ (Student)

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0

PROJECTION (π)

$\pi_{A_1, A_2, \dots, A_n}(R)$ - where A_i is attribute of a relation R

- ✱ Unary operation – take one operand
- ✱ Return specified attributes of a relation
- ✱ NO duplicates - Distinct values

Find all SID, Dept from Student

Student

SID	Name	Dept	Attn_Rate
123	Meena	CSE	1.0
211	Raju	CSE	0.6
124	Mithu	CS	0.9
153	Rita	EEE	0.8
243	Geeta	ECE	0.75

-- EQUIVALENT QUERY

SELECT DISTINCT SID, Dept FROM Student;

Solve $\pi_{\text{Dept}}(\pi_{\text{SID, Dept}}(\text{Student}))$

Note: Projection will return a set i.e: no duplicate tuple will exist

$\pi_{\text{SID, Dept}}(\text{Student})$

SID	Dept
123	CSE
211	CSE
124	CS
153	EEE
243	ECE

RENAMING (ρ)

$\rho_{R'(B_1, B_2, \dots, B_n)}(R)$ - where B_i is new attribute name of a relation R'

- ✱ Unary operation – take one operand
- ✱ Change the schema, not the instance
- ✱ Rename the result of expression E to R'
- ✱ Rename the resultant attributes to $B_1, B_2, \dots, B_n$

Other Forms:

- $\rho_S(R)$ - Changes relation name of result to S
- $\rho_{(b_1, b_2, \dots, b_n)}(R)$ - Changes attribute names of resultant

Dept

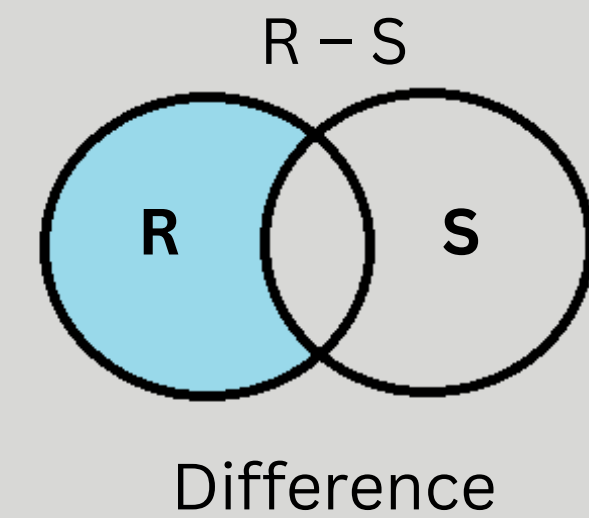
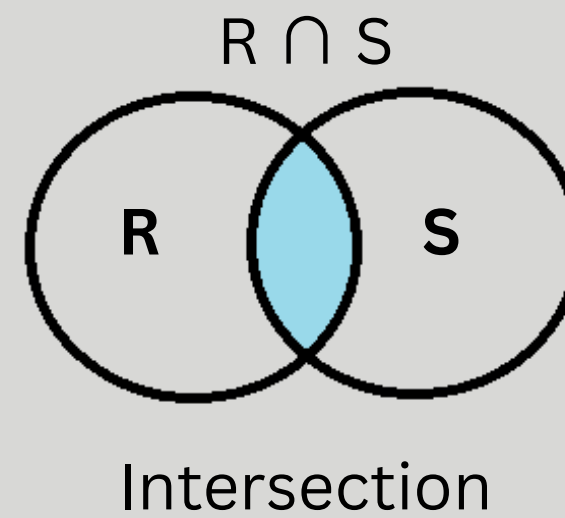
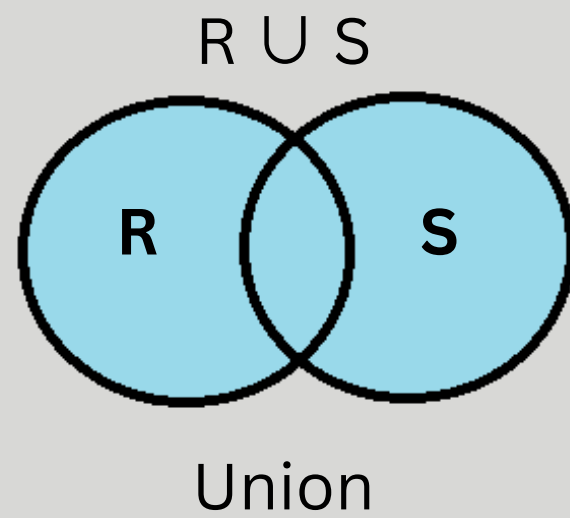
Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

$\rho_{\text{Departments}(\text{shortName}, \text{deptName})}(\text{Dept})$

Departments

ShortName	DeptName
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

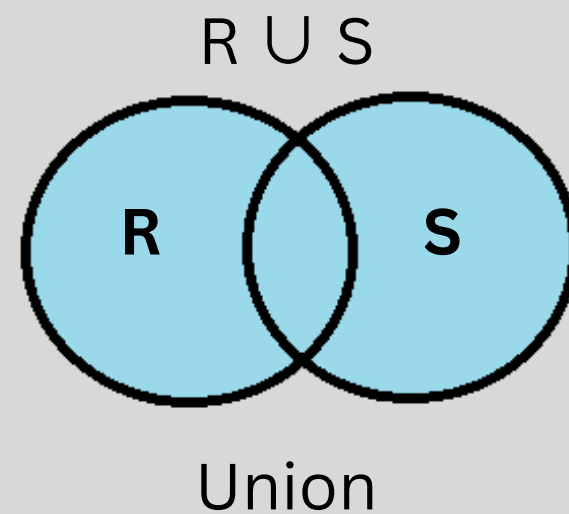
UNION, INTERSECTION, DIFFERENCE



- ✱ Binary operation – takes two operand
- ✱ Both relations **must** have the **same number of attributes** and in **same order**
- ✱ Corresponding attributes of R and S **must** be based on the **same domain/compatible data types**

UNION (U)

←
Assignment
operator



Each tuple of R and each tuple of S is put into the resultant relation

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

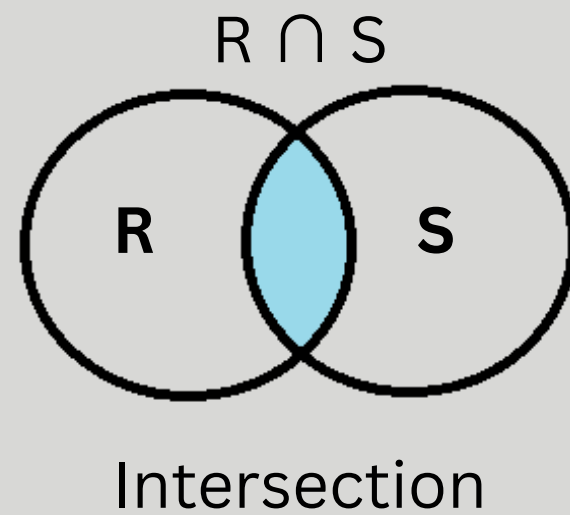
Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

All_Depts ← Old_Depts U New_Depts

All_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering
AI	Artificial Intelligence
DS	Data Science

INTERSECTION ($\cap$)



A tuple is in the resultant relation if and only if it is in both R and S

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

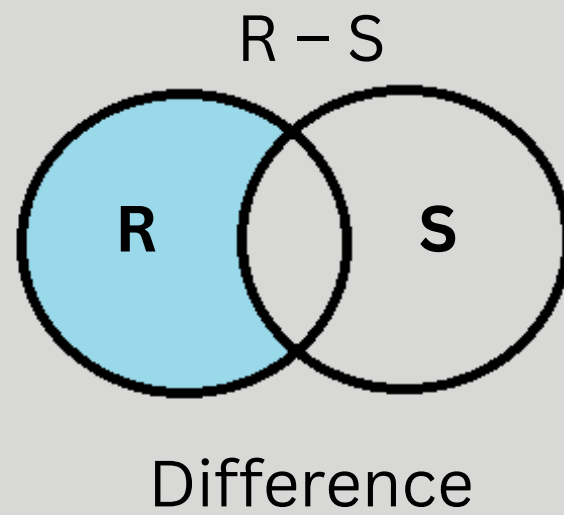
Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

Common_Depts $\leftarrow$ Old_Depts $\cap$ New_Depts

Common_Depts

Initial	Name
CS	Computer Science

DIFFERENCE (−)



A tuple is in the resultant relation if and only if it is in R but not in S

Old_Depts

Initial	Name
CSE	Computer Science and Engineering
CS	Computer Science
EEE	Electrical and Electronics Engineering

New_Depts

Initial	Name
AI	Artificial Intelligence
DS	Data Science
CS	Computer Science

Depts ← Old_Depts − New_Depts

Depts

Initial	Name
CSE	Computer Science and Engineering
EEE	Electrical and Electronics Engineering

**Solve and compare
with previous RA**

Depts ← New_Depts − Old_Depts

CARTESIAN PRODUCT ($\times$)

$R \times S$ - where $R(a_1, a_2, \dots, a_n)$ and $S(b_1, b_2, \dots, b_k)$

- ✱ Binary operation – takes two operand
- ✱ So-called “cross-product” or “product”
- ✱ Combine two relations
- ✱ Usually not meaningful when it is performed alone

$$R \times S \neq S \times R$$

StudentNames

ID	Name
123	Meena
234	Raju

StudentDepts

SID	Dept
123	CSE
234	CS

StudentNames x StudentDepts

ID	Name	SID	Dept
123	Meena	123	CSE
123	Meena	234	CS
234	Raju	123	CSE
234	Raju	234	CS

JOIN (⋈)

$R \bowtie_c S$ - where R and S has a set of attributes that are in common

- ✱ Binary operation – takes two operand
- ✱ Merge relations on the specified condition

$$R \bowtie_c S \equiv \sigma_c (R \times S)$$

StudentNames

ID	Name
123	Meena
234	Raju

StudentDepts

SID	Dept
123	CSE
234	CS

StudentNames $\bowtie_{ID = SID}$ StudentDepts

ID	Name	SID	Dept
123	Meena	123	CSE
234	Raju	234	CS

QUERY TREE (RA TREE)

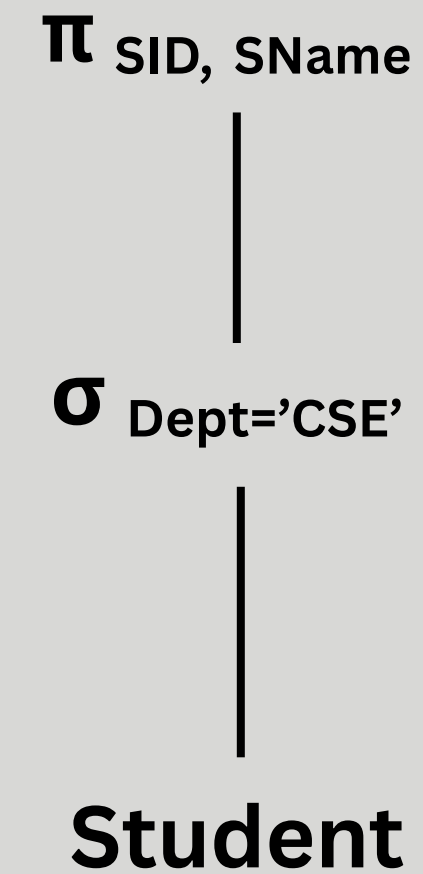
Overview

- An internal data structure to represent a query
- Estimates work involved in executing a query
- Nodes stand for *operations* (σ , π , ρ , $\bowtie$)
- Leaf Nodes represent *base relations (tables)*
- Trees can be rearranged for better performance

Example:

Query Tree for the given RA:

$\pi_{SID, SName} (\sigma_{Dept='CSE'} (Student))$



QUERY TREE (RA TREE)

Query Tree for the given RA:

$\pi_{SName, dept, assignedCourse} (ST \bowtie_{SID=SID} Student)$

$\pi_{SName, dept, assignedCourse}$

$\bowtie_{SID=SID}$

ST

Student

Query Tree for the given RA:

$\pi_{FName, Desk\#} (\sigma_{CCode='CSE370' \text{ or } CCode='CSE230'} (Facutly \bowtie_{FIN=FIN} FacultyCourses))$

$\pi_{FName, Desk\#}$

$\sigma_{CCode='CSE370' \text{ or } CCode='CSE230'}$

$\bowtie_{FIN=FIN}$

Faculty

FacultyCourses

Now Draw the RA Tree for the following RA:

$\pi_{FName, Desk\#} (\sigma_{CCode='Database Systems' \text{ or } CCode='AI'} ((Facutly \bowtie_{FIN=FIN} FacultyCourses) \bowtie_{CCode=CCode} (Course)))$

Example Problems - I

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)

ST(SID, assignedCourse)

Faculty(FIN, FName, LName, Desk#)

Course(CCode, CName)

FacultyCourses(FIN, CCode, Section)

-- FIN references Faculty(FIN), CCode references Course(CCode)

A) Write RA for the following:

1. Find students with at least 70 credits completed from 'CS' dept.
2. Find FName and Desk# of all faculties.
3. Find SID, SName of Students from 'CSE' dept.

Example - Solution (1)

A) Write RA for the following:

1. Find students with at least 70 credits completed from 'CS' dept.

Ans: $\sigma_{\text{totalCredits} \geq 70 \text{ AND Dept} = \text{'CS'}} (\text{Student})$

2. Find FName and Desk# of all faculties.

Ans: $\pi_{\text{FName, Desk\#}} (\text{Faculty})$

3. Find SID, SName of Students from 'CSE' dept.

Ans: $\pi_{\text{SID, SName}} (\sigma_{\text{Dept} = \text{'CSE'}} (\text{Student}))$

Example Problems - II

Consider the following Relational Schema:

Student (SID, SName, address, dept, totalCredits)

ST(SID, assignedCourse) -- *SID references Student(SID)*

Faculty(FIN, FName, Desk#)

Course(CCode, CName)

FacultyCourses(FIN, CCode, Section)

-- *FIN references Faculty(FIN), CCode references Course(CCode)*

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle
2. Find ST's SName, dept, assignedCourse.
3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.

Example - Solution (2)

B) Write RA for the following:

1. Rename CCode, CName columns of Course table to Code, CTitle

Ans: $\rho_{\text{Code, CTitle}}(\text{Course})$

2. Find ST's SName, dept, assignedCourse.

Ans: $\pi_{\text{SName, dept, assignedCourse}}(\text{ST} \bowtie_{\text{SID}=\text{SID}} \text{Student})$

3. Find Faculties FName and Desk# for those who are taking either CSE370 or CSE230 Courses.

Ans: $\pi_{\text{FName, Desk\#}}(\sigma_{\text{CCode}=\text{'CSE370'} \text{ or } \text{CCode}=\text{'CSE230'}}(\text{Facutly} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}))$

4. Find Faculties FName, Desk# for those who are taking either 'Database Systems' or 'AI' Courses.

Ans: $R \leftarrow ((\text{Facutly} \bowtie_{\text{FIN}=\text{FIN}} \text{FacultyCourses}) \bowtie_{\text{CCode}=\text{CCode}} (\text{Course}))$

$\pi_{\text{FName, Desk\#}}(\sigma_{\text{CCode}=\text{'Database Systems'} \text{ or } \text{CCode}=\text{'AI'}}(R))$

Note: $\leftarrow$ (Assignment operator) can be used to decompose complex queries

What Next?

LECTURE 8: INDEXING



LOADING.....

